

Machine learning approaches and databases for prediction of drug–target interaction: a survey paper

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Abstract

The task of predicting the interactions between drugs and targets plays a key role in the process of drug discovery. There is a need to develop novel and efficient prediction approaches in order to avoid costly and laborious yet not-always-deterministic experiments to determine drug–target interactions (DTIs) by experiments alone. These approaches should be capable of identifying the potential DTIs in a timely manner. In this article, we describe the data required for the task of DTI prediction followed by a comprehensive catalog consisting of machine learning methods and databases, which have been proposed and utilized to predict DTIs. The advantages and disadvantages of each set of methods are also briefly discussed. Lastly, the challenges one may face in prediction of DTI using machine learning approaches are highlighted and we conclude by shedding some lights on important future research directions.

Key words: Machine learning; drug–target interaction prediction; DTI software; DTI database

Introduction

In recent years, pharmaceutical scientists have been highly focused on novel drug development strategies that rely on knowledge about existing drugs [1–5]. Indeed, the difficulty of the drug discovery task lies in the rarity of existing drug–gene interactions [6], and a major risk is in unexpected/unintended interaction of drugs with off-target proteins, i.e. side effects [7–9]. While most of these side effects are undesired and harmful, occasionally they lead to interesting therapeutic discoveries. For instance, minoxidil was primarily developed to treat ulcers, and Sildenafil (Viagra) was developed to treat angina; however, they are currently used for treatment of hair loss and erectile dysfunction, respectively.

As such, novel drug development strategies are currently the principle focus of many pharmacologists. It has been reported that several terms such as drug repositioning, drug repurposing, drug reprofiling, drug redirecting, drug rediscovery and drug delivery have been used in the literature to describe these novel drug development strategies [3]. While various definitions have been used for these terms [3], drug repositioning usually refers to the studies that reinvestigate existing drugs that failed approval for new therapeutic indications [10], while drug repurposing suggests the application of already approved drugs and compounds to treat a different disease [11, 12].

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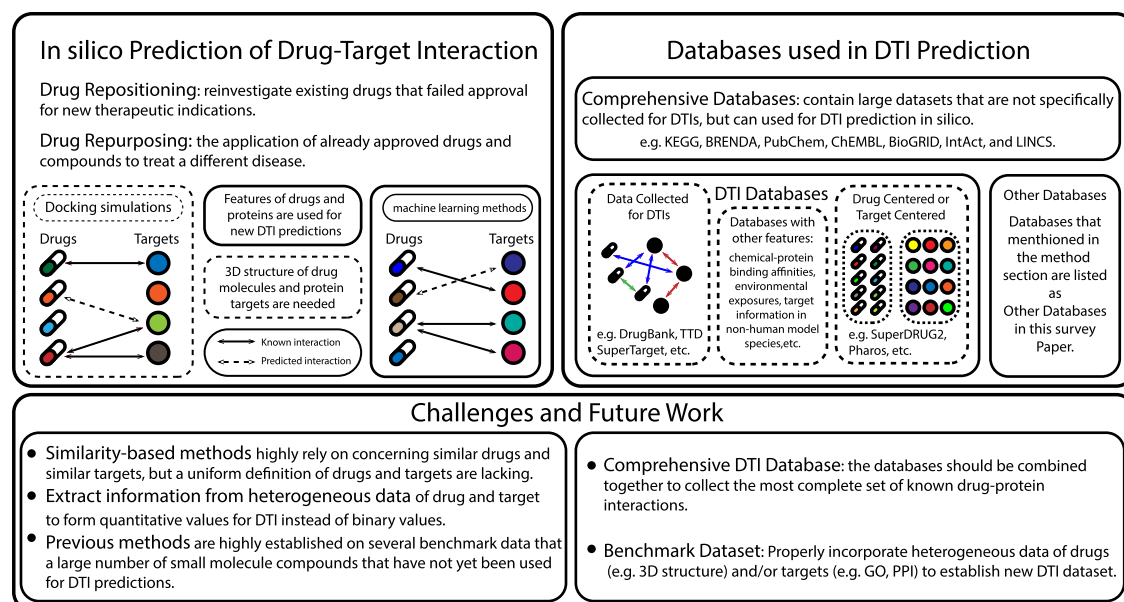


Figure 1. An overview of the present work.

A major step in the drug discovery process is to identify interactions between drugs and targets (e.g. genes), which can be reliably performed by *in vitro* experiments. In order to reduce temporal and monetary costs, *in silico* approaches are gaining more attention [2]. As such, instead of an exhausting *in vitro* search, virtual screening is initially performed and possible candidates are then experimentally verified [2]. Generally, there are two principle approaches for *in silico* prediction of drug-target interaction (DTI, also referred to as compound-protein interactions): docking simulations and machine learning methods [2]. In docking simulations, the 3D structure of drug molecules and targets are considered and potential binding sites are identified. While biologically well accepted, the docking simulation process is time-consuming [2]. Additionally, this process cannot be applied if the 3D structure of the protein is unknown [13]. For instance, for a class of proteins called G-protein-coupled receptors (GPCR), very few structures have been crystallized (orphan GPCR) [14, 15], so docking simulations cannot be applied. To tackle this issue, chemogenomics was introduced as a way to aim at mining the entire chemical space for interaction with the biological space (also referred to as genomic space), instead of considering each protein target independently from other proteins [14, 16, 17].

The aim of chemogenomics research is to relate this chemical space of possible compounds with the genomic space in order to identify potentially useful compounds such as imaging probes and drug leads [13]. Chemogenomics approaches are usually categorized as ligand based, target based and target-ligand [14, 17], all of which are based on similarities between members proteins and targets. In fact, this salient similarity-based point of view of chemogenomics allowed the machine learning approaches to be suitable for prediction of DTIs. In machine learning methods [18], knowledge about drugs, targets and already confirmed DTIs are translated into features that are used to train a predictive model, which in turn is used to predict interactions between new drugs and/or new targets. The main assumption of these studies is that if drug d is interacting with protein p , then (i) drug compounds similar to d are likely to interact with protein p , (ii) proteins similar to p are likely to interact with drug d

and (iii) drug compounds similar to d are likely to interact with proteins similar to p . The similarities between drug compounds and protein sequences are usually measured by kernels specifically designed for this purpose [19]. In practice, based on the availability of knowledge about interacting drug compounds and target proteins, the DTI prediction problem can be categorized into four classes: (i) known drug versus known target, (ii) known drug versus new target candidate, (iii) new drug candidate versus known target and (iv) new drug candidate versus new target candidate. While the ultimate goal of the machine learning methods is interaction prediction for new drug and target candidates, most of the methods in the literature are limited to the 1st three classes.

In this paper, the state of the art methods, which used machine learning methods for prediction of DTIs, are reviewed. The following studies were excluded:

- studies that do not use machine learning methods for prediction or (e.g. [20–25]).
- studies that focus on bioactivity (quantitative structure-activity relationship (SAR), proteochemometric) relationships (e.g. [26–32]).
- studies that rely on 3D structures of targets (e.g. [33–36]).
- studies that consider only the genomic space or chemical space (e.g. [4, 37–52]).
- studies that focus on gene expression for drug response (e.g. [53–58]).
- studies that only use side effect similarities or only predict side effects (e.g. [59–63]).
- studies that use disease-gene associations (e.g. [64–67]).
- studies that focus on drug-drug interactions or protein-protein interactions (PPI) (e.g. [68–72]).
- studies that use biomedical documents from which information is extracted by text mining techniques (e.g. [73]).

It is worth mentioning that the machine learning methods used in DTI prediction can be thought of as a broader problem of 'link predictions' in complex networks [74]. A section is dedicated to summarize the databases used in these studies as well. An overview of the paper is illustrated in Figure 1.

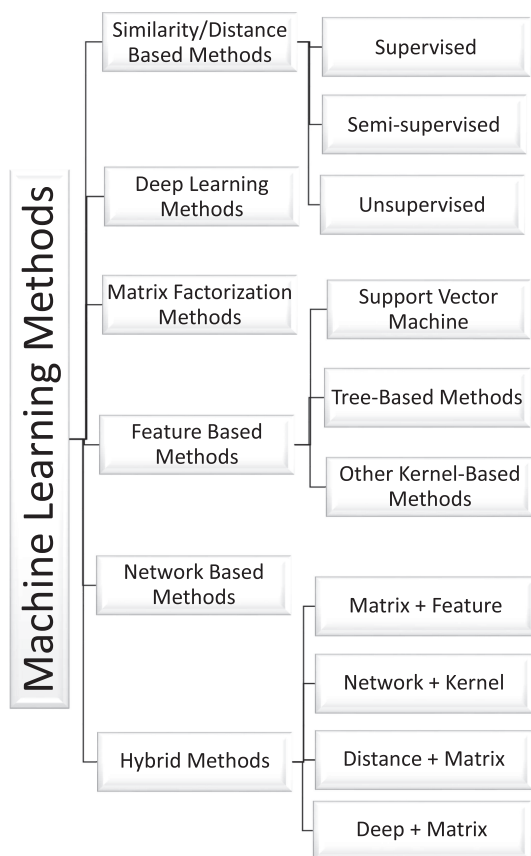


Figure 2. Machine learning methods used in DTI prediction can be categorized into six main branches. A short description of each group of methods are provided in Section 2. Here the machine learning methods are classified into similarity/distance based methods where itself consists of three subgroups. All approaches that employ kernels, trees, boosted methods, random and rotation forest, support vector machines, etc. are listed in feature-based group. Deep learning, matrix factorization and network based methods from the other three groups. Any combination of the methods listed above is considered in the category of hybrid methods.

Machine learning methods used in DTI prediction

Although all the DTI prediction frameworks that uses machine learning are summarized in this manuscript, recent methods that use matrix factorization algorithms have outperformed other methods in terms of efficiency. These methods take advantage of the recommender system approaches [75, 76], while using both chemical and genomic information is optimal for the DTI prediction problem. This problem is very similar to the famous Netflix challenge [77].

Machine learning methods used in DTI prediction date back to an early work in pharmacological DTI prediction [78]. While the focus of their work was not specifically ‘drug discovery’, they aimed at finding a ranked list of molecule ligands that bind with each orphan GPCR where due to lack of crystallized 3D structures, docking simulation could not be used [15]. Here, the machine learning approaches have been categorized into six groups (Figure 2). In the coming section, a description of each category along with a list of methods for each is provided. Moreover, advantages and disadvantages of each group of methods are briefly discussed.

Previous review papers

There have been few reviews on DTI prediction with various emphases [79–83]; however, none of these studies had a machine learning focus. For previous reviews on machine learning methods for DTI prediction, please see [84–94]. In particular, [84] is a brief review of similarity-based machine learning methods used for DTI prediction. As reported in this work, similarity-based approaches have four advantages: (i) they do not need feature extraction and feature selection, (ii) similarity measure kernels for both drugs and genes have been fully studied before, (iii) they can be easily incorporated with kernel-based learning methods such as support vector machine (SVM), (iv) they can be used to connect chemical space and the genomic space. In [85], the focus of the review is on the methods that use both drug chemical structure and target protein sequence to predict DTIs. Mousavian et al. [90] reviewed machine learning-based methods from supervised and semi-supervised perspectives. Chen et al. [91] reviewed the well-known databases, web servers and computational models used for DTI prediction. In this paper, computational approaches are divided into network-based methods and machine learning-based methods. Ezzat et al. [92] provided an ‘empirical’ overview on chemogenomic DTI prediction methods and the databases used. In their work, the chemogenomic methodologies are separated into five models: neighborhood models, bipartite local models, network diffusion models, matrix factorization models and feature-based classification models. Chen et al. [87] reviewed the machine learning methods and databases that used chemogenomic approaches of DTI prediction. As such, based on the way negative samples are handled, chemogenomic approaches are divided into two categories: (i) supervised learning methods such as similarity-based and feature-based methods, (ii) semi-supervised learning methods. Kurgan et al. [88] wrote one of the most comprehensive surveys of DTI predictions before April 2018. Sachdev et al. [93] reviewed feature-based chemogenomic approaches (excluding similarity-based chemogenomic approaches) used for DTI prediction. In this survey, feature-based methods are categorized as: (i) SVM-based methods, (ii) ensemble-based methods (methods that employ decision tree or random forest) and (iii) miscellaneous techniques (neither SVM-based nor ensemble-based). Sercinoglu et al. [94] reviewed all the available databases for drug repurposing.

Similarity/distance-based methods

The most popular group of methods used for DTI prediction incorporate drug–drug and target–target similarity measures through similarity or distance functions that are utilized to perform the prediction. These methods have been proposed and employed by several authors, mainly [13, 95–109].

Generally, the methods consist of a similarity score scheme for either drug–drug, target–target or drug–target associations based on a known pair of drug–drug and target–target similarity measures. Similarly, the similarity measure could be obtained by a distance function that defines how similar (or here ‘close’) a new drug is with respect to the known pairs. There are several ways to define the ‘nearness’ through a distance function for nearest neighbor (NN) algorithms [96, 102] among which the Euclidean distance is well known. For instance, authors in [102] employed the following definition for the NN algorithm; assuming two vector spaces (aka sample spaces) V_1 and V_2 , with the same dimension, the distance (nearness) of the two samples is denoted by $D(V_1, V_2)$,

Table 1. Similarity/distance-based methods

Abbreviations	Algorithms	Description
SITAR	Similarity-based Inference of drug-TARgets	A prediction scheme that integrates multiple drug–drug and gene–gene similarity measures to facilitate the prediction task using logistic regression [95].
SRP	Similarity-Rank-based Predictor	A lazy supervised non-parametric model using quantitative index to measure the tendency of interacting similar drugs and similar targets to predict DTIs. [97].
ECkNN /HLM	K-Nearest Neighbor Regression with Error Correction or Hubness-aware Local Models	A kNN method with an error correction method (hubness-aware regression technique) in order to alleviate the detrimental effect of bad hubs [98, 99] (with substantially different labels from those instances [100]).
NP, WP	Nearest Profile & Weighted Profile	Given a test drug candidate, it finds a known drug sharing the highest similarity with the test drug, and predict the test drug to interact with target known to interact with the nearest drug [13, 101, 102].
MDTI	MultiviewDTI	A clustering algorithm, based on spectral clustering, integrating drug data and target data from both structural and chemical views and the known DTIs [103].
STC	Super-Target Clustering	A clustering of similar targets by introducing the concept of super target to handle the missing interactions. [104].
LPLNI, LPLNI-II	Label Propagation method with Linear Neighborhood Information	A framework in which first drug–drug linear neighborhood similarity is calculated, then the manifold of drugs are taken as similarities and finally unobserved DTIs are predicted using drug–drug similarities, interaction profiles and label propagation [105].
WNN-GIP, RLS-WNN	Weighted Nearest Neighbors-GIP	A weighted NN algorithm directly incorporated into the GIP method, for constructing an interaction score profile for a new drug compound using information about known compounds [106].
BLM	Bipartite Local Models	In a bipartite graph model, predicts presence or absence of edges between drug and target using local models trained on known drugs and targets [98, 101, 107, 108].
BLM-NII	BLM with Neighbor-based Interaction-profile Inferring	An inferring integrated into the BLM method to handle the new candidate problem of pure BLM [107].
WBRDTI	Weighted Bayesian Ranking method	An improvement of BRDTI method by incorporating interaction weights for unknown DTs calculated based on known neighboring DTs [109].

where

$$D(V_1, V_2) = 1 - \frac{V_1 \cdot V_2}{\|V_1\| \|V_2\|},$$

where $(\cdot)$ and $\|\cdot\|$ denote the inner product and the Euclidean norm, respectively. One could easily verify that D is indeed a distance function satisfying the definition of the distance.

In addition to the above, the similarity/distance function could be also defined based on the pharmacological similarity of drugs and genomic similarity of protein sequences as well as the topological properties of a multipartite network of the existing drugs and protein targets [9, 110]. To this end, authors in [95] defined five drug–drug similarity measures as chemical based, ligand based, expression based, side effect based and annotation based. The main disadvantage of this group of methods lies in the fact that only a small number of drugs and their interactions are known while there exists copious unlabeled data among the datasets (see Section 3). Even though some efforts have attempted to deal with the lack of labeled data [5, 106, 107, 111, 112], the challenge has not yet been overcome. A comprehensive list of the methods proposed based on similarity/distance is provided in Table 1.

Deep learning methods

Deep learning is becoming more and more popular given its great performance in many areas, such as speech recognition, image recognition and natural language processing. Applying deep learning methods to drug discovery has been consistently increasing in recent years [113, 114].

Deep learning approaches appear to overcome certain limitations by reducing the loss of feature information in predicting DTIs. One of the drawbacks in using deep learning methods

lays in the fact that there is not always sufficient information available in order to perform deep learning methods. Recently, in order to deal with high dimensional and oftentimes noisy data in DTI predictions in general and in drug repurposing in particular, authors in [115–117] proposed and developed deep learning algorithms in the DTI's machine learning approaches.

Most of the deep learning-based DTI prediction methods consist of two major steps: generating feature vectors and then applying deep learning to known DTIs. Usually, three types of properties (i.e. biological, topological and physico-chemical information) of drugs and/or targets can be used for generating feature vectors/matrix for deep learning based DTI methods. In recently published works [116–122], methods such as deep belief neural networks [118, 119], convolutional neural networks [120, 122] and multiple layer perceptrons [121, 122] were used to establish DTI prediction programs.

In [117], instead of using a bipartite network to represent the DTI, a Tripartite Linked Network [117], derived from the existing linked open datasets in the biomedical domain [125] were used for new DTI predictions. One advantage of methods employing deep learning over the state-of-the-art feature extraction methods and SVM classifiers is the ability to mine the hidden interactions between drugs and targets.

Although all of the aforementioned deep learning methods show good performance, there is room for improvement in several aspects. First, creating robust negative datasets for supervised deep learning method is a challenging task. Most previously published deep learning based DTI prediction programs are supervised machine learning methods, so how to establish an unbiased negative DTI dataset for model fitting and testing is a key step. In addition, DTI prediction is to discover new DTIs. How to select real no-interaction drug–target pairs

Table 2. Deep learning methods

Abbreviations	Algorithms	Description
DeepDTIs	Deep Learning in predicting DTIs	A deep-learning approach utilizing DBN [123] to abstract raw input vectors and predict new DTIs between FDA approved drugs and targets [118].
DeepWalk		A deep learning similarity-based DTI prediction method based on the topology of multipartite network of the existing drugs and targets [117].
AutoDNP	Stacked Autoencoder Deep Neural Network	A deep learning computational method with an ensemble classifier using stacked Autoencoder.[116].
DeepConv-DTI	Deep learning with convolution-DTI	A deep learning method capturing local residue patterns of proteins participating in DTIs[122].
LASSO-DNN	Least absolute shrinkage and selection operator-Deep Neural Network	A deep learning method based on features extracted from the LASSO regression models fitted using the protein-specific and drug-specific features respectively [121].
DeepDTA	Deep DT Binding Affinity Prediction	A deep learning-based model using only character representations (raw sequence information) for both drugs and targets simply [120].
DeepNP	Deep Neural Representation	An interpretable end-to-end deep learning architecture to predict DTIs from low level representations [119].
DeepTrans	Deep Transcriptome data	A framework for DTI prediction based on transcriptome data in the L1000 database gathered from drug perturbation and gene knockout trials [124].

is a tricky task. Second, with more and more different types of drug/target data available, how to incorporate heterogenous data into high-dimensional features from drug and/or target for deep learning methods is also a challenge. Last but not least, deep learning methods that show great performance on the testing dataset do not mean they also can achieve great performance in real drug discovery. More details about applying deep learning in drug discovery can be found in [126]. In Table 2, a brief list of deep learning-based methods mentioned in this paper is provided.

Feature-based methods

The vast majority of machine learning methods performing DTI prediction fall into this category. It is a broad range of methods including SVM, tree-based methods and other kernel-based methods. Any pairs of drugs and targets would be represented in terms of feature vectors with certain length, often with binary labels that classify the pair vectors into two classes with positive and negative interaction. In other words, assuming feature space F where

$$F = \left\{ f := d \oplus t \mid d = [d_1, d_2, \dots, d_n] \ \& \ t = [t_1, t_2, \dots, t_m] \right\},$$

where d and t denote the target and drug feature vectors of length n and m , respectively.

Once the feature space is defined, assorted machine learning methods can be established to perform the DTI prediction task [5, 6, 9, 13, 14, 78, 89, 102, 106, 112, 127–178]. The lack of 3D structures of membrane proteins prevents extracting the main features, which otherwise would have yielded to better prediction performances. Tables 3 and 4 provides a broad list of feature-based methods along with a short description and the papers in which those methods were proposed and employed.

Matrix factorization methods

The matrix factorization methods have been shown to outperform other groups of machine learning methods in the

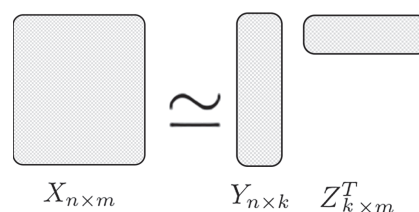


Figure 3. Matrix factorization method.

prediction of DTI. Given an interaction matrix $X_{n \times m}$,

$$X_{n \times m} = \begin{pmatrix} x_{11} & \cdots & x_{1m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nm} \end{pmatrix}$$

for $i = 1 : n$ and $j = 1 : m$, one may define

$$x_{ij} = \begin{cases} 1 & \text{if drug } d_i \text{ and target } t_j \text{ interact} \\ 0 & \text{in the absence of any known interaction} \end{cases}$$

the primarily goal in DTI prediction is to decompose matrix $X_{n \times m}$ into two matrices, $Y_{n \times k}$ and $Z_{k \times m}^T$, where $X \simeq YZ^T$ with $k < n, m$ (Figure 3). Here Z^T denotes the transposed matrix of Z . This will factorize matrix $X_{n \times m}$ into two matrices with lower orders (i.e. rank reduction), which make it easier to perform the matrix completion techniques in order to handle the missing data.

In contrast to most machine learning methods used for DTI prediction that need (2D) drug structural similarities, certain matrix factorization methods do not rely on chemical similarity or drug similarities and instead utilize collaborative filtering algorithms, among which one could name probabilistic matrix factorization (PMF) [179]. Some other methods are inspired by the idea of low-rank embedding (LRE) [180, 181] with the goal of finding a low-rank representation R of the dataset X by an optimization problem and then fixing R and minimizing the reconstruction error in the embedded space in a way that the pointwise linear reconstruction (local structure of original samples) is preserved.

In this group of methods, it is assumed that the drugs and targets are lying in the same distance space such that the distance among drugs and targets can be used to measure the

Table 3. Feature-based methods: part I

Abbreviations	Algorithms	Description
SVM, KSVM, MH-SVM	Support Vector Machine	A support vector machine constructs a hyperplane or set of hyperplanes, which can be used for prediction of presence or absence of interaction between drugs and targets [14, 78, 127–141].
BGL/KRM	Bipartite Graph Learning or Kernel Regression-based Method	In a bipartite graph model, predicts the presence or absence of edges between drug and target based on graph-based similarity to known drugs and targets in a unified Euclidean space of chemical and genomic space called pharmacological space [13, 142, 143].
NetLapRLS	RLS with kernels derived from known DTIs	The improved version of LapRLS by incorporating a new kernel established from the known DTI network [6].
PKR	Pairwise Kernel Regression	A regression model similar to KRM without requirement of any unified chemical and genomic space [9].
RF, DDR	Random Forest	A robust model against the overfitting problem of traditional statistical methods that performs more efficiently for large-scale databases [5, 131, 144, 145] (using [137, 146–150]).
iDTI-ESBoost		A prediction model for identification of DTIs using evolutionary and structural features [151].
PUDT	Positive-Unlabeled learning for DT prediction	A framework treating unknown DTI as unlabeled samples and using weighted SVM predictor [152].
GIP	RLS with Gaussian interaction profile kernel	An RLS algorithm that incorporates the topology of known DTI network as source information through GIP kernel [5, 153].
RLS	Regularized Least Square, also RLS-Kron, RLS-avg, LapRLS, KRLS, RLS-KF, KronRLS-MKL	A semi-supervised framework that incorporates known DTIs and unknown DTIs in a general-purpose learner.[6, 106, 153–158].
	SimBoost, SimBoostQuant	A non-linear method for continuous DT binding affinity prediction and an extended version SimBoostQuant, using quantile regression to estimate a prediction interval as a measure of confidence. [159].

Table 4. Feature-based methods: part II

Abbreviations	Algorithms	Description
RFDT	Rotation Forest-based DTI prediction	A computational model based on the assumptions that the protein sequences are encoded as Position Specific Scoring Matrix (PSSM) [160] descriptor and the drug molecules are encoded as fingerprint feature vector [161].
DrugRPE		A random projection ensemble approach for based on the REPTree algorithm [162] and using random projection [102, 162, 163, 163–165].
CGBVS	ChemoGenomics-Based Virtual Screening	A kernel-based state-of-the-art method using virtual screening (VS) [89] and pairwise kernel method (PKM) [14] [166].
	DASpfind	A computational DTI prediction method relying on the topological structure of the heterogeneous graph interaction model [167].
SAR	Structure-Activity Relationship method	A screening of chemical compounds method for classification problem of DTIs using protein sequences and drug topological structures [137, 168].
DVM	Discriminative Vector Machine	A classifier and a method by formulating the DTIs as an extended SAR classification problem [169] (using principal component analysis (PCA) method [170]).
EnSL	Ensemble Learning (with dimensionality reduction, or class imbalance-aware)	A framework predicts DTI based on average voting of its base classifiers: Decision Tree (EnsemDT) [171–173] (based on Singular Value Decomposition (SVD), Partial Least Squares (PLS) [174] and Laplacian Eigenmaps (LapEig) [175]), Kernel Ridge Regression (EnsemKRR), Random Forest (EnsemRF) [112], stacked (EnsemSTACK) [176], DrugE-Rank [177].
BE-DTI	Bagging-based Ensemble method	A bagging-based ensemble framework that involves dimensionality reduction and active learning [178].

strength of their interactions. Therefore, both drugs and targets can be embedded in a common low-dimensional subspace with some constraints.

Although this group of methods has been shown to be more reliable than the others, rapid growth in the quantity and variety of data related to a certain drug and/or a target far exceeds the capacity of matrix-based data representations and many current analysis algorithms. A solution to this issue has been proposed in Section 4. In Table 5, the matrix factorization methods and the

paper(s) in which they are proposed, developed and employed are listed.

Network-based methods

The network-based methods refer to those that utilize graph-based techniques in order to perform the task of DTI prediction (Figure 4). Among the methods is network-based inference (NBI) for DTI prediction, which is among the simplest yet most reliable

Table 5. Matrix factorization methods

Abbreviations	Algorithms	Description
MSCMF	Multiple Similarities one-Class Matrix Factorization	An approach to approximate the input DTI matrix by two low-rank matrices, which share the same feature space and are generated by the weighted similarity matrices of drugs and those of targets, respectively [182] using [183–186].
NRLMF	Neighborhood Regularized Logistic Matrix Factorization	A mode that integrates logistic matrix factorization with neighborhood regularization for DTI prediction [187].
PMF	Probabilistic Matrix Factorization	A collaborative filtering method that decomposes the DT bipartite connectivity matrix as a product of two matrices of latent variables that will be used for prediction, irrespective of the drug or target similarities [179].
DLGRMC	Dual Laplacian Graph Regularized Matrix Completion	An optimization framework for low-rank approximation of interaction matrix based on matrix completion in which drug similarity and target similarity are used as dual Laplacian graph regularization term [188].
GRMF-WGRMF	Graph Regularized Matrix Factorization and Weighted GRMF	Two manifold learners for extracting low-dimensional non-linear manifolds of DTI bipartite graph [189].
Pseudo-SMR	Pseudo Substitution Matrix Representation	An extension to SAR classification problem[137], employing a python package called <i>scikit-learn</i> for machine learning to implement Extremely Randomized Tree (ER-Tree) introduced in [190, 191].
BRDTI	Bayesian Ranking method	A method based on Bayesian Personalized Ranking matrix factorization (BPR) that incorporates target bias and content alignment for drug and target similarities [2, 99, 192].
LRE, SLRE, MLRE	Low Rank Embedding	An algorithm of finding a low-rank representation (by optimization problem) and fixing and minimizing the reconstruction error in the embedded space in a way that the pointwise linear reconstruction (local structure of original samples) is preserved [181]. LRE for an arbitrary view (structure or chemical) is called SLRE and for multiview is called MLRE [180].
VB-MK-LMF	Variational Bayesian Multiple Kernel Logistic Matrix Factorization	A method integrating multiple kernel learning, weighted observations, graph Laplacian regularization and explicit modeling of probabilities of binary DTIs [193].
KBMF, KBMF2K	Kernelized Bayesian Matrix Factorization	A method for factorizing the interaction score matrix in terms of kernel matrices (similarity matrices), which can be used as DTI predictors for new drugs and protein KBMF2K [194].

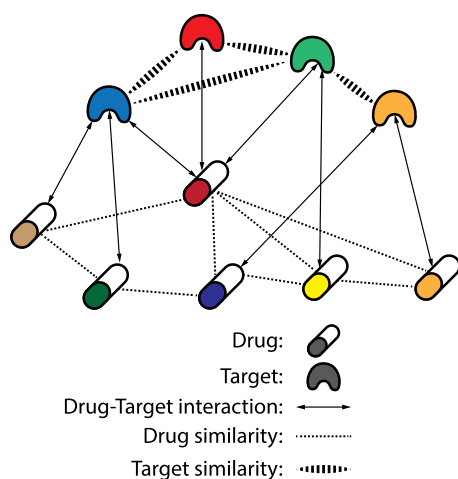


Figure 4. Drug–target interaction heterogeneous network.

inference methods and uses only DT bipartite network topology similarity [195].

Moreover, in certain methods three networks of protein–protein similarity, drug–drug similarity and known DTIs are integrated into a heterogeneous network and assumed similar drugs often target similar proteins [196, 203]. A two-layer undirected graphical representation of the network could also be adopted in order to train to predict direct DTIs (usually caused by protein–ligand binding), indirect DTIs and drug mode of actions

(binding interaction, activation interaction and inhibition interaction) in addition to performing the DTI prediction task. A pertinent example is proposed in [204] using Restricted Boltzmann Machine (RBM) [123]. A list of network-based methods with a short description for each method is provided in Table 6.

Hybrid methods

Hybrid methods refer to all the approaches in which any combination of the feature-based, matrix factorization, deep learning and network-based methods are exploited. This can extend the capability of a prediction algorithm by integrating different sets of information. The hybrid methods in general serve two purposes; they address the problems of unknown interaction in DTIs as well as taking the most advantage of machine learning methods, simultaneously. For instance, authors in [177] proposed a method integrating feature-based and similarity-based machine learning approaches [205, 206]. The hybrid methods performed superior to other state-of-the-art methods by optimizing the feature extraction process by extracting the complex hidden features of drugs and targets [134, 144, 172, 173, 182, 197, 201, 207, 208]. Integrating two machine learning methods in DTI prediction often has a leverage in terms of results as they fully exploit the potential of two methods, simultaneously. However, one should be able to deal with the high complexity (either computational or operational) caused by integrating two groups of methods. A short description of such methods are listed in Table 7.

Table 6. Network-based methods

Abbreviations	Algorithms	Description
NBI	Network-Based Inference	A method based on DT bipartite network topology similarity [195].
NRWRH	Network-based Random Walk with Restart on the Heterogeneous network	A method based on the framework of RWR to infer potential DTIs on a bipartite graph network [196].
NetCBP	Network-Consistency-based Prediction Method	A semi-supervised inference method, utilizing both labeled and unlabeled data [111].
DTINet		A computational network integration pipeline for DTI prediction [197].
IN-RWR	inter/intra-network RWR or Co-rank	Two network prediction methods based on Co-rank algorithm that involves RWR on bipartite graph [198].
	NormMullInf	A method based on collaborative filtering that incorporates multiple available data sources related to drugs and targets can improve DTI prediction performance [199] using robust PCA [200].
NRLMF β	Beta-distribution-rescored NRLMF	An improved NRLMF algorithm that rescores the score of NRLMF as the expected value of the β -distribution, which is determined based on interaction information and NRLMF score. [201].
RWR	Random Walk with Restart	A method that requires a matrix inversion and provides a good relevance score between two nodes in a weighted graph of DTIs [202].

Table 7. Hybrid methods

Abbreviations	Algorithms	Description
DT-Hybrid	Domain tuned-hybrid	An extended NBI technique that incorporates domain-based knowledge such as drug similarities and target similarities [209] (also look [195, 210, 210, 211, 211??] for extension of the capability of recommender systems).
KMDR	Kernel Matrix Dimension Reduction	A framework for construction of link similarity matrix from kernel matrix and feature transformation for DTI prediction [208].
MGRNNM, DGRMC	Multi Graph Regularized Nuclear Norm Minimization	A computational method that adds multiple drug-graph and target-graph Laplacian regularization terms to the standard matrix completion framework to predict DTIs [212, 213].
WLMN	Weisfeiler-Lehman Neural Machine	An algorithm for extraction of the adjacency matrix that represents the interactions between potential drugs and targets [214].
PDTPS	Predicting Drug Targets with Protein Sequence	A framework based on Relevance Vector Machine that integrates Bi-gram probabilities, PSSM and PCA [215].
	L_1 -regularized Classifier	A regularized classifiers over the tensor product space of DT pairs for extracting informative and biologically meaningful features for DTI prediction [216].
RBM	Restricted Boltzmann Machine	A two-layer undirected graphical model to represent a multidimensional DTI network and encode different types of DTIs [204].
LRF-DTI	Lasso-based Random Forest method	A method of DTI prediction based on Lasso dimensionality reduction and random forest predictor [144].
COSINE	COLD Start INteractions	A statistical dual-regularized, one-class collaborative filtering method [217] framework and a corresponding computational method for multi-target virtual screening using one-class collaborative filtering technique that can employ either logistic or linear factorization [218].
DMF	Deep Matrix Factorization	A deep learning approach in the context of recommendation systems to extract the non-linearity of latent variables [219] (DMF was originally introduced in [220] as a deep learning method in the context of recommendation systems to extract the non-linearity of latent variables).
CoDe-DTI	Collaborative DEep learning-based DTI predictor	A method using both PMF and a denoising autoencoder [221].

Software and packages

Sakakibara et al. [222] developed a web service called Comprehensive Predictor of Interactions between Chemical compounds And Target proteins based on their previous works [127, 129] that uses SVM as the DTI predictor. This server seems to be no longer available.

Cao et al. [223] developed a Python package called PyDPI based on Random Forest [150] that integrates chemoinformatics, bioinformatics, proteochemometrics and chemogenomics for DTI prediction. The proposed framework involves

the selection of molecular features and uses predefined dictionaries for classification. This package can be used to construct web-based servers and provides an interface for databases such as Kyoto Encyclopedia of Genes and Genomes (KEGG), PubChem, Drugbank and Uniprot. The same group in the same year [224] also developed a web-based server called PreDPI-Ki (which seems to be no longer available) based on a random forest predictor that takes binding affinities of DT pairs into account in order to better predict interactions.

Table 8. DTI databases

Database	Latest updates	No. of targets	No. of drugs/compounds	No. of int.	Predicted DTIs	Search fun.
ChEMBL	Dec 2018	12 482	1 879 206	15 504 603	✗	✓
ChemProt 3.0	Dec 2015	>20 000	>1 700 000	-	✗	✓
DGIdb 3.0	Nov 2017	41 100	9495	29 783	✗	✓
DrugBank	Apr 2019	5175	13 338	26 932	✗	✓
GtoPdb	Jun 2019	2926	9718	>50 000	✗	✓
IntAct	May 2019	102 508	10 849	593 007	✗	✓
KEGG	May 2019	-	-	-	✗	✓
LINCS	2016	1469	41 847	-	✗	✗
PROMISCUOUS	May 2011	6548	5258	23 702	✗	✓
STITCH	Jan 2016	>9 600 000	>430 000	-	✓	✓
SuperTarget	Oct 2011	>6000	>196 000	>330 000	✗	✓
TTD	Sep 2017	3101	34 019	-	✗	✓

Xiao et al. [225] established a web server called iGPCR-drug, which is accessible at iGPCR-drug. Moreover, they developed a sequence-based classifier also called iGPCR-drug. In the predictor, the drug compound is formulated by a 2D fingerprint via a 256D vector, GPCRs by the pseudo amino acid composition [226] generated with the gray model theory and the prediction engine is operated by the fuzzy K-nearest neighbor (KNN) classification method [227]. The authors validated their method with the jackknife test [228].

Yamanishi et al. [229] designed a web server called DINIES (DTI network inference engine based on supervised analysis) for predicting DTI using various types of biological data such as chemical structures, protein domain and drug side effects (note that studies that primarily focused on side effect are excluded in this paper [59–62]) and three supervised algorithms (BGL [13, 143], BLM [101] and pairwise kernel regression [9]). This is due to the work by Scheiber et al. [230] that enables the calculation of correlation between any drug compound and pharmacological effects in chemical space. While the training can be performed using KEGG DRUG database, the principle advantage of their web server is the flexibility of the input data, as long as it's represented a similarity matrix or gene/drug profile.

Seal et al. [231] developed a standalone R and Shiny package called Netpredictor based on Random Walk with Restart (NRWRH) [196, 202] and NBI [195, 209] to predict any missing links between drugs, proteins and drug-proteins in any unipartite or bipartite. The main advantage of this package is the friendly user interface that is provided by package installation.

Hao et al. [232] review, compare and reimplemented five state-of-the-art methods (BLM [101], KronRLS-MKL [158], DT-Hybrid [209], the proposed method by Shi et al. [104] and DNILMF [233]) and published the source codes in R.

Databases used in DTIPrediction

To support the above methods, many drug-related databases have been established. These databases contain different types of drug-related information and are critical resources for DTI predictions in silico. In this paper, we review all popular used databases associated with this topic. Based on the content of these databases, we classify them into four categories, DTI databases, drug-centered or target centered databases, drug-target binding affinity databases and supporting databases.

DTI databases

DTI databases are established for collecting DTIs and other related information. In this paper, we list 11 databases in this category. Within these databases, some are not directly proposed as 'DTI' databases, but the data contained can be used for DTI research. For example, KEGG is an extensive database that covers many types of biological data from genes/proteins to biological pathways and human diseases. In KEGG [234], two subdatabases, KEGGDRUG [235] and KEGGBRITE [236] contain data that can be used for DTI predictions. ChEMBL [237–239] is also not specifically a 'drug-target' database and it was established based on collecting bioactive compounds. However, combined with targets and other related biological information, this database can also be used in drug-target repositioning and repurposing. Similar to ChEMBL [237–239], IntAct [240] is a database that contains molecular interactions and can be used for drug research. LINCS is different from the aforementioned two databases. This data portal contains biochemistry data that aims to understand changes in gene expression and cellular processes that are caused by different perturbing agents. Many of the perturbing agents used in LINCS are drugs, so this is also a great data source for DTI research. Other databases included in this group are SuperTarget [241], Guide to PHARMACOLOGY (GtoPdb) [240], DrugBank [242–246], Therapeutic Targets Database (TTD) [247], STITCH [248–252], ChemProt 3.0 [253] and DGIdb 3.0 [254]. The general information for these databases is summarized in Table 8.

ChEMBL

The data stored in the ChEMBL database [237–239] were manually extracted from published literatures. This database was published by European Molecular Biology Laboratory (EMBL)-European Bioinformatics Institute in 2002. Since the latest update in 2018, this database contains more than 1.9 million chemical compounds. Within these compounds, over 10 thousand drugs and more than 12 thousand targets are included in ChEMBL.

ChemProt 3.0

ChemProt [253, 255, 256] was proposed as a disease chemical biology database that integrated data from multiple chemical-protein annotation databases and disease-associated PPI. The first release of ChemProt was in 2011, which collected data from eight public databases, i.e. ChEMBL [238], BindingDB [257], PDSP Ki database [258], DrugBank [244], PharmGKB [259], PubChem bioassay [260], CTD [261] and STITCH [248] and two commercial

databases, WOMBAT and WOMBAT-PK [262]. The second update of ChemProt was in 2012 integrated therapeutic effects and adverse drug reactions into the 2.0 version. The latest update (version 3.0) was released in 2015. The third version updated the disease chemical biology data. In addition, several computational methods, such as network biology based enrichment analysis, were also incorporated.

DGIdb 3.0

The first release (in 2013) of DGIdb integrated 13 data sources that cover information in disease-related human genes, drugs, drug interactions and potential druggability [263, 264]. The latest update of DGIdb was in 2017 and in total 30 data sources are included in the 3.0 version [254]. Six new data sources were added and nine of the previous data sources were updated.

DrugBank

DrugBank [242–246] is one of the most popular databases and has been widely used as a drug reference resource. This database was first released in 2006. As a database both in bioinformatics and cheminformatics, DrugBank contains detailed drug data with comprehensive drug target information. The DTI relationships in DrugBank were originally collected from textbooks, published articles and other electronic databases. All data can be freely downloaded from DrugBank.

GtoPdb

This database was established by the International Union of Basic and Clinical Pharmacology/British Pharmacological Society. The GtoPdb [240] contains the ligand–activity–target relationships data that were collected from pharmacological and medicine chemistry literature.

IntAct

IntAct [265] is an open source database of molecular interactions populated by data from literature and other data sources. In total, 11 molecular interaction databases (including IntAct) were incorporated into IntAct including AgBase [266–269], MINT [270–273], UniProt [274][41], I2D [275], MBINFO, MatrixDB [276], Molecular Connections, InnateDN [277], IMEx [278] and GOA.

KEGG

KEGG is a comprehensive database that provides many types of knowledge about genes and genomes [234, 235]. The whole database can be summarized in four major categories. The first one is systems information, contains three databases: KEGG PATHWAY, KEGG BRITE, and KEGG MODULE. The second category contain genomic information. In this group, four databases are included: KEGG ORTHOLOGY, KEGG GENOME, KEGG GENES and KEGG SSDB. The third category holds the chemical information. Five databases are in this category: KEGG COMPOUND, KEGG GLYCAN, KEGG REACTION, KEGG RCLASS and KEGG ENZYME. The last category is health information that covers four databases: KEGG DISEASE, KEGG DRUG, KEGG DGROUP and KEGG ENVIRON. The KEGG DGROUP database contains information regarding drug interaction networks including DTIs, drug metabolism and indirect interactions with enzymes and target genes.

LINCS

The LINCS program aims to establish a network-based landscape to describe how different perturbing agents influence

cellular processes. In total, there are 398 datasets collected in the LINCS database including fluorescence imaging, ELISA and ATAC-seq data, etc. The majority datasets (177 datasets) in LINCS are KINOMEScan kinase-small molecule binding assays. This assay is used to measure binding interactions between test compounds.

PROMISCUOUS

PROMISCUOUS was established in 2011 and proposed as a database for network-based drug repositioning. This database contains three different types of data: drugs, proteins and side effects. The protein data are extracted from UniProt and incorporated with the 3D structure information from Protein Data Bank (PDB). Drugs and side effects are extracted and incorporated from SuperDrug and SIDER, respectively. In addition to DTIs and drug side effects linkages, PROMISCUOUS also includes data on drug–drug similarities and PPI.

STITCH

STITCH [248–252] is a database that stores information for interactions between proteins and small molecules. The interaction data are collected from predicted results, other databases (e.g. PubChem [279]), and literature. The first release of STITCH was in 2008.

SuperTarget

SuperTarget [241] is a database that covers DTI information with drug metabolism, pathways and Gene Ontology (GO) terms. Medical indications and adverse drug effects are also included in this database. The DTIs information in this database were extracted starting with text mining from 15 million public literature listed in PubMed. Also, potential drug–target relations were also extracted from Medline. Furthermore, the relationships of DTIs from other databases (i.e. DrugBank [244], KEGG [234], PDB [280], SuperLigands [281] and TTD [282]) were also used to obtain any missed DTIs that were not included from the previous two strategies.

DTT provides therapeutic proteins, nucleic acid targets and corresponding drug information [247]. This database was first described in 2002. The data in TTD was mainly collected from literature. Other databases that contains DTIs information (e.g. KEGG) were also cross-linked to TTD.

Drug-centered or target-centered databases

In this category, six databases are included. They are BRENDA [283], PubChem [279], SuperDRUG2 [284], DrugCentral [285, 286], PDID [287], Pharos [288] and ECDrug [289].

Among these databases, SuperDRUG2 and DrugCentral are proposed as ‘drug-centered’ databases. Since PubChem is a database established on collecting millions of chemical compounds, in this paper, we also list this one as a ‘drug-centered’ database. PDID and Pharos are classified as ‘target-centered’ databases. We also included BRENDA as a ‘target database’. The huge amount of enzymes and related ligands stored in BRENDA can be used as targets in DTI research. In addition, we also list ECDrug here as a target-centered database. Different from the aforementioned ones, this database contains target information in non-human model species. Relative information can be found in Table 9.

BRENDA

BRENDA [283, 290] is a comprehensive enzyme database that was first established in 1987. This database contains 784 000

Table 9. Drug-centered or Target-centered databases

Database	Latest updates	Type	No. of targets	No. of drugs/Compounds	Predicted DTIs
BRENDA	Jan 2019	Target centered	>84 000	>205 000	✗
DrugCentral	Apr,2019	Drug centered	-	4543	✗
ECODrug	Oct 2017	Target centered	-	-	✗
PDID	Apr 2015	Target centered	3746	51	✓
Pharos	Nov 2018	Target centered	20 244	130 166	✗
PubChem	Mar 2019	Drug centered	79 622	96 157 016	✗
SuperDRUG2	Mar 2018	Drug centered	4456	4605	✓

enzymes and their corresponding enzyme–ligand related information. All data collected in this database was manually evaluated and extracted from ~140 000 literature references based on the Enzyme Commission (EC) classification system of the International Union of Biochemistry and Molecular Biology. All compounds related to enzyme catalyzed reactions are labeled as ‘ligands’ in BRENDA, such as substrates, products, activators, inhibitors and cofactors. In total, about 205 000 enzyme ligands were collected and stored in the associated ligand database. Users can search the ligand database through the search box on the home page. BRENDA also provides download functionality for users to download all BRENDA data.

DrugCentral

DrugCentral is a comprehensive database that focuses on drug collection [285, 286]. This database was released in 2016 and contains approved active pharmaceutical ingredients (drugs) from FDA and other regulatory agencies. For each drug, structure information, bioactivity and regulatory records, as well as pharmacologic actions and indications were incorporated. In this database, all drugs are simply classified into three categories, small molecule active ingredients, biological active ingredients and others.

ECODrug

In drug discovery research, non-human model species are important in that they are used for drug testing. ECODrug [289] is a database that contains DTI data for 640 eukaryotic species. The data stored in ECODrug can help researchers investigate the conservation of human drug targets across species. The drug information and drug targets are from previous research [291] and DrugBank [244].

PDID

PDID [287] was released in 2014 and covers all known protein–drug interactions and predicted protein–drug interactions for the entire structural human proteome. The known interactions were extracted from DrugBank [244], BindingDB [257] and PDB [280]. The predictions were made by using three different softwares (i.e. ILbind [292], SMAP [45] and eFindSite [293, 294]).

Pharos

Pharos [288] is a platform that was established for presenting the data in the Target Central Resource Database (TCRD). TCRD is a comprehensive database that was initially developed for discovering new druggable proteins.

The data stored in TCRD came from many different sources. It includes biomedical literature, expression data, disease and phenotype association data, bioactivity data, DTI data and databases from Harmonizome [295].

PubChem

PubChem [279, 296] stores the information of chemical substances and corresponding biological actives. This database consists of three sub-databases: Substance, Compound and BioAssay. Substance is the primary repository to store chemical information provided from individual data contributors. The Compound database contains the unique chemical structures extracted from the Substance database. All biological related data of these chemical substance data are saved in the BioAssay database.

SuperDRUG2

SuperDRUG2 [284] is proposed as a one-stop data source that offers all crucial features of approved and marketed drugs. The drug items in SuperDRUG2 are classified into two categories: small molecules and biological/other drugs. Several public resources like US FDA, CFDA and EMA, etc. were used for drug collections. Drug target information in SuperDRUG2 was extracted from DrugBank [244], TTD [247] and ChEMBL [238]. Besides these drugs and targets information, SuperDRUG2 also provides 2D and 3D structure information of small molecule drugs, drug side effects, drug–drug interactions and drug pharmacokinetic parameters.

Binding affinity databases

In this category, BindingDB [257, 297–299], PDBbind [300] and PDSP Ki [301] are included. All of them contain the data on chemical–protein binding affinities. BindingDB is mainly focused on collection of binding affinity data between drugs (drug-like molecules) and target proteins. PDBbind is established based on binding affinity measurements of biomolecular complexes from PDB. PDSP Ki is similar to BindingDB, which also contains a large number of binding affinity data on DTIs. Table 10 shows the relative information of these three databases.

BindingDB

BindingDB [257, 297–299] is a repository that contains experimental protein–small molecule interaction information. All of these data were extracted from scientific literature and US patents. In addition, other databases (e.g. ChEMBL [238], PubChem [296], etc.) are also linked with BindingDB.

PDBbind

PDBbind [300] was first released in 2004 and the purpose of this database is to bridge the gap between protein structural information and energetic properties. The data stored in PDBbind were classified by the biomolecular complex data from PDB. Then, the binding affinity data were collected from the

Table 10. Binding affinity databases

Database	Latest updates	No. of targets	No. of drugs/compounds	No. of DTI	No. of TTI
BindingDB	May 2019	7269	733198	1651120	-
PDBBind	Jan 2018	-	-	16276	3312
PDSP Ki	2019	-	-	-	-

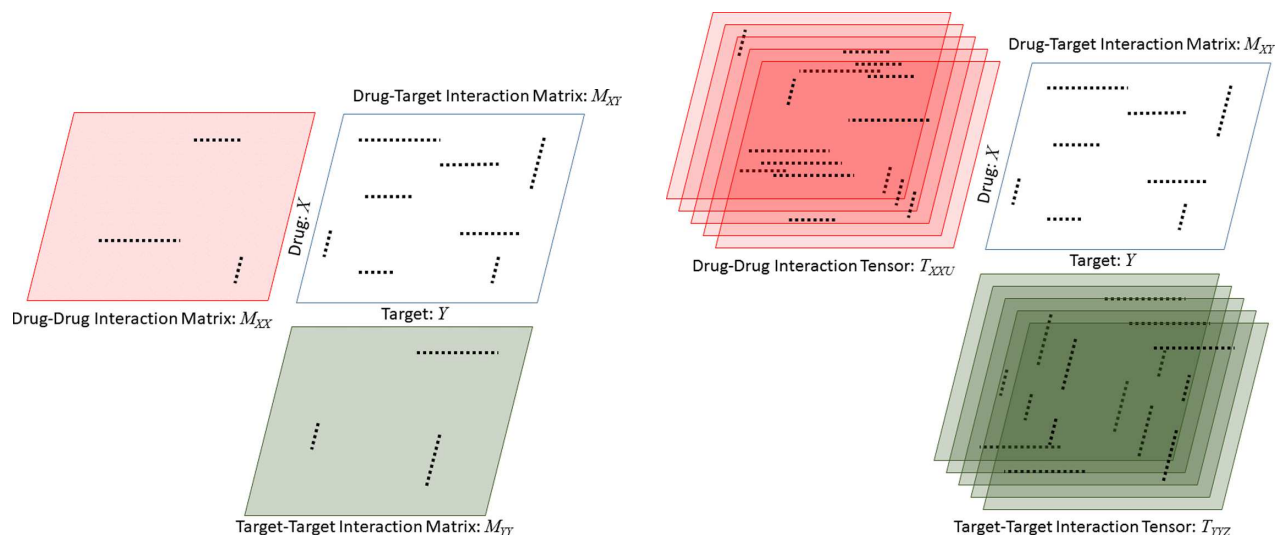


Figure 5. Coupled matrix-matrix versus coupled tensor-matrix.

associated literature on PDB. PDBbind has regular updates with the growth of PDB database.

PDSP Ki

PDSP Ki [301] is a public database that stored binding affinities data of drugs/chemical compounds for four different types of proteins, i.e. receptors, neurotransmitter transporters, ion channels and enzymes. This database was developed and maintained by University of North Carolina at Chapel Hill. Search function for both drugs and targets are provided.

DTI database challenges and future work

The challenges in making reliable predictions of DTI can be classified into two main categories: the challenges concerning the databases and those concerning computations. Oftentimes, one may overcome the computational difficulties using different prediction methods depending on the nature of the problem. However, major challenges arise due to the source of the databases. Here, we provide some challenges of the first type, also discussed by authors in [88, 92], followed by some suggestions on how to deal with the challenges in future work.

Database challenges and future work

Almost all the methods used in DTI prediction, particularly similarity-based methods, heavily rely on assertions concerning similar drugs and similar targets, the type of database used for the prediction plays a significant role. In terms of databases, lacking a uniform definition of drugs and targets as well as a consistent way of calling and identifying compounds and biomolecules, overlapping with at least one other source in the pool, adopting different identifiers to represent drug and targets

are among the main challenges [88, 92]. Additionally, incorporating heterogeneous data in a database is another challenge to be pointed out. Not all the drugs and targets included in a database have 3D structures and GO/PPI sequences, respectively, which makes similarity scores. As a consequence, the resulting data could vary even if the same literature is used.

Future predictions should rely on more comprehensive internal databases, which would require a significant effort to map and curate data across the sources that utilize different ways to define, name and identify the drugs and targets. From the data perspective, there is an issue of datasets being of a binary nature; i.e. given an interaction matrix $X_{n \times m}$, for $i = 1, \dots, n$ and $j = 1, \dots, m$, one may define

$$x_{ij} = \begin{cases} 1 & \text{if drug } d_i \text{ and target } t_j \text{ interact} \\ 0 & \text{in the absence of any known interaction.} \end{cases}$$

This causes a significant problem. Some of the 0's in $X_{n \times m}$ may be interactions that are yet undiscovered, which may throw off the training process for the different classifiers. Another point is that in reality DT pairs have binding affinities that vary over a spectrum (interactions are not binary on/off). One suggestion to overcome this challenge is to utilize datasets with continuous values representing DT binding affinities. This have been previously proposed by authors in [5, 131, 153, 302, 303]. Our suggestion is to replace each x_{ij} with continuous-valued parameters. Based on the probability of interaction, one may define $x_{ij} = \mu$ where $\mu \in [0, 1]$. 0, as it should, indicates no interaction while 1 denotes complete interaction. Any number within (0, 1) represents the probability that drug d_i and target t_j interact.

Table 11. The summary of all algorithms and databases

Study	Algorithm	Database
Bock et al. [78]	SVM	PDSP Ki, Swiss-Prot (UniProt), Ligand.Info, ExpASy
Faulon et al. [130]	SVM	PTC, KEGG, DrugBank
Nagamine et al. [129]	SVM	DrugBank, UniProt, PubChem, PDSP Ki, GLIDA
Nagamine et al. [127]	SVM	DrugBank, UniProt, NIST05, CE-MS
Wassermann et al. [128]	SVM	MEROPS, CutDB, SCOP, MDDR, PDB, BindingDB
Jacob et al. [14]	SVM	KEGG BRITE
Cao et al. [137]	SVM	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Liu et al. [138]	SVM	DrugBank, Matador, STITCH, PubChem, SIDER
Mousavian et al. [136]	SVM	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Shen et al. [135]	SVM	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Ding et al. [134]	SVM	KEGG BRITE, BRENDA, SuperTarget, DrugBank, ChEMBL, Matador
Yamanishi et al. [143]	BGL	KEGG DRUG, KEGG LIGAND, KEGG GENES, KEGG BRITE, BRENDA, SuperTarget, DrugBank, JAPIC
Yamanishi et al. [13]	BGL or KRM, NN	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Bleakley et al. [101]	BLM, KRM, NN	KEGG BRITE, BRENDA, SuperTarget, DrugBank
He et al. [102]	NN	KEGG BRITE, KEGG LIGAND, KEGG GENES, BRENDA, SuperTarget, DrugBank
Xia et al. [6]	LaRLS, NetLapRLS	KEGG LIGAND, KEGG GENES
Van Laarhoven et al. [153]	GIP, RLS	KEGG BRITE, KEGG LIGAND, KEGG GENES, BRENDA, SuperTarget, DrugBank
Perlman et al. [95]	SITAR	KEGG DRUG, DrugBank, DCDB, SuperTarget, REACTOME, CTD
Takarabe et al. [9]	PKR	AERS, SIDER, JAPIC, KEGG DRUG, KEGG GENES
Gonen [194]	KBMF	KEGG BRITE, KEGG LIGAND, KEGG GENES, BRENDA, SuperTarget, DrugBank
Cheng et al. [195]	NBI, TBSI, DBSI	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Chen et al. [196]	NRWRH	KEGG LIGAND, KEGG BRITE, BRENDA, SuperTarget, DrugBank
Mei et al. [107]	BLM-NII	KEGG BRITE, KEGG LIGAND, KEGG GENES, BRENDA, SuperTarget, DrugBank
Yu et al. [131]	SVM, RF	DrugBank
Tabei et al. [216]	L_1 -regularized	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Wang et al. [204]	RBM	MATADOR, STITCH
Zheng et al. [182]	MSCMF	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Van Laarhoven et al. [106]	WNN-GIP	KEGG BRITE, KEGG LIGAND, KEGG GENES, BRENDA, SuperTarget, DrugBank
Cobanoglu et al. [179]	PMF	DrugBank
Alaimo et al. [209]	DT-Hybrid	KEGG BRITE, BRENDA, SuperTarget, DrugBank ([195])
Chen et al. [111]	NetCBP	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Tabei et al. [139]	MH-SVM	STITCH, PubChem, UniProt, PFAM
Pahikkala et al. [5]	RF, RLS	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Niu et al. [112]	EnsemRF	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Bharadwaja [156]	KRLS	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Kuang et al. [155]	RLS-Kron	DrugBank, KEGG LIGAND, UniProt
Peng et al. [199]	NormMulInf	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Zhang [176]	EnsemSTACK	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Seal et al. [202]	RWR	DrugBank, ChEMBL
Shi et al. [104]	Super-Target Clustering	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Shi et al. [97]	SRP	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Lan et al. [152]	PUDT	KEGG BRITE, BRENDA, SuperTarget, DrugBank, KEGG LIGAND
Liu et al. [187]	NRLMF	KEGG BRITE, BRENDA, SuperTarget, DrugBank, ChEMBL, KEGG LIGAND
Ezzat et al. [171]	EnsemDT	DrugBank
Ba-alawi et al. [167]	DASpfind	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Yuan et al. [177]	DrugE-Rank	DrugBank
Hao et al. [157]	RLS-KF	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Nascimento et al. [158]	KronRLS-MKL	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Lim et al. [218]	COSINE	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Buza et al. [98]	ECkNN, HLM	KEGG BRITE, BRENDA, SuperTarget, DrugBank, Kinase, KEGG GENES
Peska et al. [2]	BPR, BRDTI	KEGG BRITE, BRENDA, SuperTarget, DrugBank, Kinase
Meng et al. [215]	PDTPS	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Zhang et al. [105]	LPLNI	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Ezzat et al. [172, 173]	EnsemDT, EnsemKRR	DrugBank ([171]), KEGG BRITE, BRENDA, SuperTarget, DrugBank
Ezzat et al. [189]	GRMF	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Kuang et al. [208]	KMDR	DrugBank, KEGG LIGAND, UniProt

(Continued)

Table 11. Continued

Study	Algorithm	Database
Olayan et al. [145]	RF (DDR)	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Zhang et al. [103]	MultiviewDTI	DrugBank
Li et al. [180]	LRE	DrugBank, KEGG
Wen et al. [118]	DeepDTIs	DrugBank
Luo et al. [197]	DTINet	DrugBank, HPRD
Zong et al. [117]	DeepWalk	DrugBank
He et al. [159]	SimBoost, SimBoostQuant	Kinome Datasets in [307, 308]
Li et al. [169]	DVM	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Zhang et al. [164]	DrugRPE	KEGG BRITE, KEGG LIGAND, KEGG GENES, BRENDA, SuperTarget, DrugBank ([102])
Rayhan et al. [151]	iDTI-ESBoost	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Hao et al. [233]	DNILMF	KEGG BRITE, BRENDA, SuperTarget, DrugBank, KEGG GENES, KEGG DRUG, KEGG COMPOUND
Ohue et al. [166]	CGBVS	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Wang et al. [188]	DLGRMC	KEGG BRITE, BRENDA, SuperTarget, DrugBank, KEGG LIGAND, KEGG GENES
Sharma et al. [178]	BE-DTI'	DrugBank, KEGG
Shi et al. [109]	WBRDTI	KEGG BRITE, BRENDA, SuperTarget, DrugBank, Kinase
Huang et al. [198]	IN-RWR, Co-rank	DrugBank, DGIdb, TTD
Shi et al. [144]	LRF-DTI	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Kadiyala [214]	WLMN	KEGG BRITE, KEGG LIGAND, KEGG GENES, BRENDA, SuperTarget, DrugBank ([106])
Manoochehri et al. [219]	DMF	KEGG BRITE, KEGG LIGAND, KEGG GENES, BRENDA, SuperTarget, DrugBank ([106])
Mongia et al. [212, 213]	MGRNNM, DGRMC	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Wan et al. [207]	NeoDTI	DrugBank, HPRD ([197])
Wang et al. [116]	AutoDNP	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Huang et al. [191]	Pseudo-SMR	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Wang et al. [161]	RFDI	KEGG BRITE, BRENDA, SuperTarget, DrugBank
Ban et al. [201]	NRLMF β	KEGG BRITE, BRENDA, SuperTarget, DrugBank, KEGG LIGAND, KEGG GENES
Bolgar et al. [193]	VB-MK-LMF	KEGG DRUG, KEGG BRITE, BRENDA, SuperTarget, DrugBank
Lee et al. [122]	DeepConv-DTI	DrugBank 4.0 [243], KEGG, International Union of Basic and Clinical Pharmacology (IUPHAR) [309]
You et al. [121]	LASSO-DNN	Drugbank
Özgür et al. [120]	DeepDTA	Kinase [308], KIBA [310]
Gao et al. [119]	DeepNP	BindingDB [257]
Xie et al. [124]	DeepTrans	DrugBank

The trend of using such continuous-valued datasets may eventually catch on as it is more useful and more meaningful, in the sense that it represents the reality better than the binary datasets that have been used in the majority of previous work in DTI prediction. The main challenge, however, lies in the fact that to date, there is a large number of small molecule compounds that have not yet been used as drugs and for the majority of them, their interaction profiles with proteins are still unknown.

Future work on DTI predictions could be categorized in two main approaches. Modifications and suggestions toward the databases in general seem inescapable. On the one hand, the databases should be combined together to collect the most complete set of known drug-protein interactions. On the other hand, the sources should regularly be updated and disseminated, which results in improvements and completeness. A larger number of source databases should be integrated to derive the internal database.

DTI prediction method challenges and future work

Future research should focus on methods that combine multiple similarities. The ensemble-based models that combine multiple types of similarities are likely to provide more accurate results than the methods that use one similarity. For instance,

repurposed drugs have been identified via retrospective clinical analysis (e.g. reviewing side effects), pharmacological analysis or simply serendipity. Given the surprisingly successful early examples (repurposing minoxidil from hypertension to hair loss, sildenafil from angina to erectile dysfunction and thalidomide from morning sickness to multiple myeloma), research is now focusing on how best to adopt a more comprehensive, systematic approach. In addition, a great amount of work is invested to identify molecular drivers of disease development, progression and treatment resistance, providing many candidate targets for drugs across the spectrum of human disease. However, a majority of these molecular drivers have no known drug to target them. Thus, a comprehensive, improved methodology for predicting DTIs would have great benefit. Due to challenges listed in Section 4.1, current knowledge of which cellular molecules are targeted by a drug is scarce and is derived from various, sometimes complementary sources.

As per the formulation of the problem, appropriate representation of datasets seems crucial for gaining insight and effectiveness in DTI predictions. In Big Data applications it is common that data is sparse (mostly zeros) and partially missing. Missing data imputation, especially in the context of sparse, noisy data, is therefore a central problem. To infer the missing entries from the known ones, reasonable assumptions should be made based

on commonly observed challenges in the structure of data. Considering matrix factorization methods in predicting DTIs, a common situation is a matrix with missing entries (such as the famous Netflix problem.) Under the assumption that the completed matrix has low rank, the low-rank matrix completion problem is NP hard and highly non-convex [304], but there are various algorithms that work under certain assumptions of the data. One approach to low rank matrix completion is to use the nuclear norm as a convex relaxation of the matrix rank, and use semidefinite programming to find a completion that minimizes the nuclear norm (see [305, 306]). Although the low-rank matrix completion problem does not depend on any metric, most approaches utilize some kind of metric (such as the nuclear norm, the Euclidean metric or an ℓ_p -norm). Such approaches may perform well in completion of certain matrix types but do not cover all types of matrices. Moreover, the structure of the data may be more complicated than a matrix with dimension $d = 2$. To this end, it is our belief that coupled matrices and tensors are very powerful tools to visualize DT data while maintaining the structural information. For $d \geq 3$, such a dataset is a tensor (a multidimensional array) of order d . Tensors are ubiquitous in Big Data. The importance of using tensors in Big Data is illustrated by the fact that they preserve the structure of the data and allow more effective data analysis by incorporating the structure throughout the process. An illustration of coupled matrix-matrix versus coupled tensor-matrix completion is shown in Figure 5.

Summary of materials and methodologies

Table 11 summarizes all the methods we reviewed in this paper along with the databases.

Key Points

- **Machine learning:** To our best knowledge, this manuscript is the first which provides a comprehensive list of all the machine learning methods that have been proposed, developed and employed to carry out the task of DTI prediction. A classification of these methods along with advantages and disadvantages of each class of method have been provided.
- **DTI software and packages:** A list and a short description of all the key software used in DTI predictions is provided. This could help future research, based on their approach to the problem, by helping researchers decide which software and packages suit their problem the best.
- **DTI databases:** One of the main challenges in the prediction of DTIs is the fact that not all the interactions between drugs and targets are known. In fact, the number of unknown interactions far exceeds the number of known interactions. As a partial solution, a comprehensive list of all databases along with the most recent update dates and the focus are provided.

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